FactProbe: A Benchmark for Measuring Hallucination in Large Language Models

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Abstract

FactProbe is a benchmark for Hallucination Detection in a Large Language Model. It contains 5,000 questions with verified answers and annotations marking the exact spans where a response introduces unsupported content. The benchmark also defines an automatic hallucination-rate metric that counts responses containing at least one unsupported span. We use FactProbe to compare three generic model sizes and RetGen, a retrieval-augmented system. Larger systems hallucinate less often, but every size remains above a meaningful error floor. RetGen reduces the rate by supplying evidence, yet it does not eliminate unsupported claims. FactProbe is designed to make these failures measurable rather than hiding them behind aggregate answer scores. Its annotations support both system ranking and error analysis across factual, numerical, and attributional statements.

Keywords

Hallucination, FactProbe, Claim evaluation, Language model reliability

1. Introduction

Fluent answers can still contain unsupported claims, creating a need for direct Hallucination Detection. Existing answer accuracy scores may miss partially correct responses, so FactProbe evaluates the text span where a claim departs from verified evidence. We are motivated by retrieval-augmented generation [1], which was expected to fix hallucination but only partially does so. This paper introduces a benchmark and an evaluation suite rather than a new generator.

2. Method

FactProbe samples 5,000 questions and obtains a verified reference answer for each. Annotators mark hallucination spans at token level and assign a coarse error category. The benchmark score is the fraction of responses containing one or more marked unsupported spans. We evaluate small, medium, and large generic models using the same prompt and decoding budget, then evaluate RetGen with its retrieved context. To avoid conflating refusal with correctness, empty answers are scored separately and excluded from the span-rate denominator only when the model explicitly declines to answer. The release includes question identifiers, reference evidence, span annotations, and scripts for macro-averaging rates. This makes Hallucination Detection reproducible for any Large Language Model that can return plain text.

3. Experiments

The small, medium, and large generic models hallucinate on 28.4%, 21.7%, and 16.3% of FactProbe questions, respectively. RetGen lowers the rate to 12.8%, but remains above zero. Larger models are safer on average, while retrieval helps across all sizes. The most common failures are confident additions to otherwise correct answers and unsupported numerical details.

System	Hallucination rate
Generic small	28.4%
Generic medium	21.7%
Generic large	16.3%
RetGen	12.8%

4. Conclusion

FactProbe makes Hallucination Detection concrete by pairing verified answers with span-level annotations. Scale helps and RetGen reduces the error rate, but neither removes the problem. The benchmark therefore supports continued work on verification, evidence selection, and transparent uncertainty for every Large Language Model.

References

[1] Vik, A. (2021). Retrieve-then-Generate: Augmenting Language Models with Retrieved Passages for Knowledge-Intensive Tasks. Conference on Natural Language Systems (CNLS).